

## 5. The Dirac Field.

Scalar fields  $\phi(x)$  :

$$\phi(x) \rightarrow \phi'(x) = \phi(\Lambda^{-1}x) \quad [\text{under Lorentz transformation}]$$

$$x^\mu \rightarrow (x')^\mu = \Lambda^\mu{}_\nu x^\nu$$

Quantization of such fields give spin-0 particle

Vector fields  $A^\mu(x)$  :

$$A^\mu(x) \rightarrow \Lambda^\mu{}_\nu A^\nu(\Lambda^{-1}x)$$

In general, a field can transform as

$$\phi^a(x) \rightarrow D[\Lambda]^a{}_m \phi^m(\Lambda^{-1}x)$$

$D[\Lambda]$  : a representation of the Lorentz group :  $D[\Lambda_1] D[\Lambda_2] = D[\Lambda_1 \Lambda_2]$

$$D[\Lambda^{-1}] = D[\Lambda]^{-1}, \quad D[1] = 1$$

Q: How do we find the different representations?

A: infinitesimal transformations of the Lorentz group.

$$\Lambda^\mu{}_\nu = \delta^\mu{}_\nu + \omega^\mu{}_\nu$$

↑  
infinitesimal  $\omega$

$$\text{Then} \quad \Lambda^\mu{}_\sigma g^\sigma{}_\rho \Lambda^\rho{}_\nu = g^\mu{}_\nu \quad \longrightarrow \quad \omega^\mu{}_\nu + \omega^\nu{}_\mu = 0 \quad [\text{antisymmetric}]$$

An antisymmetric  $4 \times 4$  matrix has 6 independent components :

6 transformations of the Lorentz group: 3 rotations & 3 boosts

One possible basis :  $(\gamma^A)^\mu{}_\nu \quad (A = 1, 2, \dots, 6)$

In practice, replacing the single index  $A$  with  $[\rho\sigma] \quad (\rho, \sigma = 0, \dots, 3)$

↑  
antisymmetric e.g.  $\mu^{01} = -\mu^{10}$

With this notation, we can write a basis as

$$(\gamma^{\rho\sigma})^\mu{}_\nu = i[g^{\rho\mu}g^\sigma{}_\nu - g^{\sigma\mu}g^\rho{}_\nu] \quad \left\{ \begin{array}{l} \rho\sigma: \text{index of basis elements} \\ \mu\nu: \text{index of the } 4 \times 4 \text{ matrix} \end{array} \right.$$

Lower one index, we have

$$(J^{\sigma\tau})^\mu{}_\nu = i[g^{\mu\sigma}\delta^\tau{}_\nu - g^{\sigma\mu}\delta^\tau{}_\nu]$$

$\uparrow$   
 no longer antisymmetric

E.g.

$$(J^{01})^\mu{}_\nu = -\left(\begin{array}{ccc|ccc} 0 & i & & & & \\ i & 0 & & & & \\ \hdashline & & & 0_{2 \times 2} & & \\ & & & & 0_{2 \times 2} & \end{array}\right)$$

$\uparrow$   
 real, symmetric ; boost in  $x^1$

$$(J^{12}) = \begin{pmatrix} 0 & 0 & 0 & 0 \\ 0 & 0 & -i & 0 \\ 0 & i & 0 & 0 \\ 0 & 0 & 0 & 0 \end{pmatrix}$$

$\uparrow$   
 real, antisymmetric ; rotation in  $(x^1, x^2)$

Any  $\omega^\mu{}_\nu$  can be expressed as a linear combination of  $J^{\sigma\tau}$

$$\omega^\mu{}_\nu = -\frac{i}{2} \Omega_{\sigma\tau} (J^{\sigma\tau})^\mu{}_\nu$$

$\uparrow$                        $\nwarrow$   
 6 numbers              the generators of the Lorentz transformation

The generators obey the Lorentz Lie algebra

$$[J^{\sigma\tau}, J^{\mu\nu}] = i(g^{\sigma\mu}J^{\tau\nu} - g^{\sigma\nu}J^{\tau\mu} + g^{\tau\mu}J^{\sigma\nu} - g^{\tau\nu}J^{\sigma\mu})$$

Then a finite Lorentz transformation is

$$\Lambda = \exp\left(-\frac{i}{2} \Omega_{\sigma\tau} J^{\sigma\tau}\right)$$

## 5.1 The spinor representation.

Clifford algebra,  $\{\gamma^\mu, \gamma^\nu\} = \gamma^\mu \gamma^\nu + \gamma^\nu \gamma^\mu = 2g^{\mu\nu} \mathbb{1}$   
 $\uparrow$   
 $\gamma^\mu$  with  $\mu = 0, 1, 2, 3$   $\uparrow$   
the unit matrix

$$\Rightarrow \gamma^\mu \gamma^\nu = -\gamma^\nu \gamma^\mu \quad \text{when } \mu \neq \nu$$

$$(\gamma^0)^2 = 1, \quad (\gamma^i)^2 = -1, \quad i = 1, 2, 3$$

The simplest representation is  $4 \times 4$  matrices.

e.g. Weyl or chiral representation:

$$\gamma^0 = \begin{pmatrix} 0 & 1_{2 \times 2} \\ 1_{2 \times 2} & 0 \end{pmatrix} \quad \gamma^i = \begin{pmatrix} 0 & \sigma^i \\ -\sigma^i & 0 \end{pmatrix}$$

Pauli matrices:  $\sigma^1 = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$ ,  $\sigma^2 = \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix}$ ,  $\sigma^3 = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}$

with  $\{\sigma^i, \sigma^j\} = 2\delta^{ij}$

Relation of Clifford algebra and the Lorentz group.

Considering the commutator of two  $\gamma^\mu$ .

$$S^{\rho\sigma} = \frac{i}{4} [\gamma^\rho, \gamma^\sigma] = \begin{cases} 0 & \rho = \sigma \\ \frac{i}{2} \gamma^\rho \gamma^\sigma & \rho \neq \sigma \end{cases} = \frac{i}{2} \gamma^\rho \gamma^\sigma - \frac{i}{2} \gamma^\sigma \gamma^\rho$$

NOTE:  $[\gamma^{\mu\nu}, \gamma^\rho] = i(\gamma^\mu g^{\nu\rho} - \gamma^\nu g^{\mu\rho})$

$[\gamma^{\mu\nu}, S^{\rho\sigma}] = i(g^{\nu\rho} \gamma^{\mu\sigma} - g^{\mu\rho} \gamma^{\nu\sigma} + g^{\mu\sigma} \gamma^{\nu\rho} - g^{\nu\sigma} \gamma^{\mu\rho})$

## Spinors :

Introduce the Dirac spinor field  $\psi^\alpha(x)$ . Under Lorentz transformation

$$\psi^\alpha(x) \rightarrow S[\Lambda]^\alpha_\beta \psi^\beta(\Lambda^{-1}x)$$

with  $\Lambda = \exp\left(-\frac{i}{2} \Omega_{\mu\nu} J^{\mu\nu}\right)$

$$S[\Lambda] = \exp\left(-\frac{i}{2} \Omega_{\mu\nu} S^{\mu\nu}\right)$$

### Rotations

$$S^{ij} = \frac{i}{2} \begin{pmatrix} 0 & \sigma^i \\ -\sigma^i & 0 \end{pmatrix} \begin{pmatrix} 0 & \sigma^j \\ -\sigma^j & 0 \end{pmatrix} = \frac{1}{2} \varepsilon^{ijk} \begin{pmatrix} \sigma^k & 0 \\ 0 & \sigma^k \end{pmatrix} \equiv \frac{1}{2} \varepsilon^{ijk} I^k \quad [i \neq j]$$

the rotation parameters:  $\Omega_{ij} = \varepsilon_{ijk} y^k$  [e.g.  $\Omega_{12} = y^3$ ]

Then the rotation matrix:

$$S[\Lambda] = \exp\left(-\frac{i}{2} \Omega_{\mu\nu} S^{\mu\nu}\right) = \begin{pmatrix} e^{-i \vec{y} \cdot \vec{\sigma}/2} & 0 \\ 0 & e^{-i \vec{y} \cdot \vec{\sigma}/2} \end{pmatrix}$$

e.g. a rotation by  $2\pi$  about  $x^3$ -axis,  $\vec{y} = (0, 0, 2\pi)$

then  $S[\Lambda] = \begin{pmatrix} e^{-i\pi\sigma^3} & \\ & e^{-i\pi\sigma^3} \end{pmatrix} = -1_{4 \times 4}$

Therefore, under a  $2\pi$  rotation  $\psi^\alpha(x) \rightarrow -\psi^\alpha(\Lambda^{-1}x)$

Boost :

$$S^{0i} = \frac{i}{2} \sigma^0 \sigma^i = \frac{i}{2} \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} \begin{pmatrix} 0 & \sigma^i \\ -\sigma^i & 0 \end{pmatrix} = \frac{i}{2} \begin{pmatrix} -\sigma^i & 0 \\ 0 & \sigma^i \end{pmatrix}$$

the boost parameters:  $\Omega_{0i} = -\Omega_{i0} = \beta_i$

$$S[\Lambda] = \exp\left(-i \Omega_{0i} S^{0i}\right) = \begin{pmatrix} e^{-\vec{\beta} \cdot \vec{\sigma}/2} & \\ & e^{\vec{\beta} \cdot \vec{\sigma}/2} \end{pmatrix}$$



A Lorentz invariant action:

Start in a naive way: define  $\psi^\dagger(x) = (\psi^*)^T(x)$

under Lorentz transformation

$$\psi(x) \rightarrow S[\Lambda] \psi(\Lambda^{-1}x)$$

$$\psi^\dagger(x) \rightarrow \psi^\dagger(\Lambda^{-1}x) S[\Lambda]^\dagger$$

$$\text{Then } \psi^\dagger(x) \psi(x) \rightarrow \psi^\dagger(\Lambda^{-1}x) \underbrace{S[\Lambda]^\dagger S[\Lambda]} \psi(\Lambda^{-1}x)$$

Since the representation is not unitary,  $S[\Lambda]^\dagger S[\Lambda] \neq 1$

$$\text{NOTE: } S[\Lambda]^\dagger = \exp\left[\frac{i}{2} \Omega_{\rho\sigma} (S^{\rho\sigma})^\dagger\right] = \gamma^0 S[\Lambda]^{-1} \gamma^0.$$

$$\begin{aligned} \text{Derivation: } \gamma^0 \gamma^\mu \gamma^0 &= (\gamma^{0\mu} - \gamma^\mu \gamma^0) \gamma^0 \\ &= \gamma^{0\mu} \gamma^0 - \gamma^\mu (\gamma^0)^2 \quad \leftarrow (\gamma^0)^2 = 1 \\ &= \begin{cases} \gamma^0 & \mu = 0 \\ -\gamma^\mu & \mu = 1, 2, 3 \end{cases} \\ &= (\gamma^\mu)^\dagger \end{aligned}$$

$$\begin{aligned} (S^{\mu\nu})^\dagger &= -\frac{i}{4} [(\gamma^\mu \gamma^\nu)^\dagger - (\gamma^\nu \gamma^\mu)^\dagger] \\ &= -\frac{i}{4} [(\gamma^\nu)^\dagger (\gamma^\mu)^\dagger - (\gamma^\mu)^\dagger (\gamma^\nu)^\dagger] \\ &= -\frac{i}{4} [\gamma^0 \gamma^\nu (\gamma^0)^2 \gamma^\mu \gamma^0 - \dots] \\ &= \gamma^0 S^{\mu\nu} \gamma^0 \end{aligned}$$

$$\begin{aligned} S[\Lambda]^\dagger &= \exp\left[\frac{i}{2} \Omega_{\rho\sigma} (S^{\rho\sigma})^\dagger\right] \\ &= \exp\left[-\frac{i}{2} \Omega_{\rho\sigma} \gamma^0 S^{\rho\sigma} \gamma^0\right] \\ &= \sum_{n=0}^{\infty} \frac{1}{n!} \left(-i \frac{\Omega_{\rho\sigma}}{2} \gamma^0 S^{\rho\sigma} \gamma^0\right)^n \\ &= \gamma^0 S[\Lambda]^{-1} \gamma^0 \end{aligned}$$

Define the Dirac adjoint

$$\bar{\psi}(x) = \psi^\dagger(x) \gamma^0$$

- NOTE:
- $\bar{\psi}\psi$  is a Lorentz scalar
  - $\bar{\psi}\gamma^\mu\psi$  is a Lorentz vector

Derivation: ①  $\bar{\psi}(x) \psi(x) = \psi^\dagger(x) \gamma^0 \psi(x)$

$$\begin{aligned} &\rightarrow \psi^\dagger(\Lambda^{-1}x) S[\Lambda]^\dagger \gamma^0 S[\Lambda] \psi(\Lambda^{-1}x) \\ &= \psi^\dagger(\Lambda^{-1}x) \gamma^0 S[\Lambda]^{-1} (\gamma^0)^2 S[\Lambda] \psi(\Lambda^{-1}x) \\ &= \bar{\psi}(\Lambda^{-1}x) S[\Lambda]^{-1} S[\Lambda] \psi(\Lambda^{-1}x) \\ &= \bar{\psi}(\Lambda^{-1}x) \psi(\Lambda^{-1}x) \end{aligned}$$

②  $\bar{\psi}(x) \gamma^\mu \psi(x) \rightarrow \bar{\psi}(\Lambda^{-1}x) S[\Lambda]^{-1} \gamma^\mu S[\Lambda] \psi(x)$

We work infinitesimally.

$$\Lambda = \exp\left(-\frac{i}{2} \Omega_{\rho\sigma} \mathcal{M}^{\rho\sigma}\right) = 1 - \frac{i}{2} \Omega_{\rho\sigma} \mathcal{M}^{\rho\sigma} + \dots$$

$$S[\Lambda] = \exp\left(-\frac{i}{2} \Omega_{\rho\sigma} S^{\rho\sigma}\right) = 1 - \frac{i}{2} \Omega_{\rho\sigma} S^{\rho\sigma} + \dots$$

then  $S[\Lambda]^{-1} \gamma^\mu S[\Lambda] = \Lambda^\mu{}_\nu \gamma^\nu$

$$\Rightarrow \left(1 + \frac{i}{2} \Omega_{\rho\sigma} S^{\rho\sigma}\right) \gamma^\mu \left(1 - \frac{i}{2} \Omega_{\rho\sigma} S^{\rho\sigma}\right) = \gamma^\nu - \frac{i}{2} \Omega_{\rho\sigma} (\mathcal{M}^{\rho\sigma})^\mu{}_\nu \gamma^\nu$$

$$\Rightarrow \gamma^\mu + \frac{i}{2} \Omega_{\rho\sigma} S^{\rho\sigma} \gamma^\mu - \frac{i}{2} \Omega_{\rho\sigma} \gamma^\mu S^{\rho\sigma}$$

$$= \gamma^\mu + \frac{i}{2} \Omega_{\rho\sigma} [S^{\rho\sigma}, \gamma^\mu]$$

$$\downarrow [S^{\rho\sigma}, \gamma^\mu] = i(\gamma^\rho g^{\sigma\mu} - \gamma^\sigma g^{\rho\mu})$$

$$= \gamma^\mu + \frac{1}{2} \Omega_{\rho\sigma} (g^{\rho\mu} \gamma^\sigma - g^{\sigma\mu} \gamma^\rho)$$

$$\downarrow (\mathcal{M}^{\rho\sigma})^\mu{}_\nu \gamma^\nu = i(g^{\rho\mu} \gamma^\sigma - g^{\sigma\mu} \gamma^\rho)$$

$$= i(g^{\rho\mu} \gamma^\sigma - g^{\sigma\mu} \gamma^\rho)$$

$$= \gamma^\mu - \frac{i}{2} \Omega_{\rho\sigma} (\mathcal{M}^{\rho\sigma})^\mu{}_\nu \gamma^\nu$$

$$\simeq \Lambda^\mu{}_\nu \gamma^\nu$$

$$\Rightarrow S[\Lambda]^{-1} \gamma^\mu S[\Lambda] = \Lambda^\mu{}_\nu \gamma^\nu$$

Dirac action:

$$S = \int d^4x \bar{\psi}(x) (i \gamma^\mu \partial_\mu - m) \psi(x)$$

After quantization this theory describes particles and anti-particles of mass  $|m|$  and spin  $1/2$ .

$\Rightarrow$  Dirac equation

$$(i \gamma^\mu \partial_\mu - m) \psi = 0$$

$$(i \partial_\mu \bar{\psi} \gamma^\mu + m \bar{\psi}) = 0$$

The Dirac equation mixes up different components of  $\psi$  through  $\gamma^\mu$ . However, each individual component solves Klein-Gordon equation.

$$\begin{aligned} & (i \gamma^\nu \partial_\nu + m) (i \gamma^\mu \partial_\mu - m) \psi = 0 \\ & = (- \gamma^\nu \gamma^\mu \partial_\nu \partial_\mu - m^2) \psi \\ & = \left[ -\frac{1}{2} (\gamma^\nu \gamma^\mu + \gamma^\mu \gamma^\nu) \partial_\mu \partial_\nu - m^2 \right] \psi \\ & = - (\partial_\mu \partial^\mu + m^2) \psi \end{aligned}$$

Slash notation:  $A_\mu \gamma^\mu = \not{A} \quad \Rightarrow \quad (i \not{\partial} - m) \psi = 0$

$$\bar{\psi} (i \overleftarrow{\not{\partial}} - m) = 0$$

## Chiral Spinors.

$$\text{Chiral representation: } \gamma^0 = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, \quad \gamma^i = \begin{pmatrix} 0 & \sigma^i \\ -\sigma^i & 0 \end{pmatrix}$$

In this representation,  $S[\Lambda_{\text{rot}}]$  &  $S[\Lambda_{\text{boost}}]$  are block diagonal,

$$S[\Lambda_{\text{rot}}] = \begin{pmatrix} e^{-i\vec{\phi} \cdot \vec{\sigma}/2} & 0 \\ 0 & e^{-i\vec{\phi} \cdot \vec{\sigma}/2} \end{pmatrix} \quad S[\Lambda_{\text{boost}}] = \begin{pmatrix} e^{-\vec{\kappa} \cdot \vec{\sigma}/2} & 0 \\ 0 & e^{\vec{\kappa} \cdot \vec{\sigma}/2} \end{pmatrix}$$

In the chiral representation,  $\psi = \begin{pmatrix} \psi_L \\ \psi_R \end{pmatrix}$  [ $u_{\pm}$ : Weyl spinors]

In group theory language,  $\psi_L$  is in the  $(\frac{1}{2}, 0)$  representation of the Lorentz group,  $\psi_R$  is in  $(0, \frac{1}{2})$  representation. The Dirac spinor  $\psi$  lies in the  $(\frac{1}{2}, 0) \oplus (0, \frac{1}{2})$  representation.

Using Weyl spinors, we have

$$\begin{aligned} \mathcal{L} &= \bar{\psi} (i\not{\partial} - m) \psi \\ &= \psi^\dagger \gamma^0 (i\not{\partial} - m) \psi \\ &= (\psi_L^\dagger, \psi_R^\dagger) \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} \left[ i\gamma^\mu \begin{pmatrix} 0 & \sigma^\mu \\ \bar{\sigma}^\mu & 0 \end{pmatrix} - m \right] \begin{pmatrix} \psi_L \\ \psi_R \end{pmatrix} \quad \left[ \begin{array}{l} \sigma^\mu \equiv (1, \sigma^i), \\ \bar{\sigma}^\mu \equiv (1, -\sigma^i) \end{array} \right] \\ &= i\psi_L^\dagger \bar{\sigma}^\mu \partial_\mu \psi_L + i\psi_R^\dagger \sigma^\mu \partial_\mu \psi_R - m\psi_R^\dagger \psi_L - m\psi_L^\dagger \psi_R \end{aligned}$$

A massive fermion needs both  $\psi_L$  and  $\psi_R$ .

A massless fermion can be described by  $u_+$  or  $u_-$  alone.

$$\left. \begin{aligned} i\bar{\sigma}^\mu \partial_\mu \psi_L &= 0 \\ i\sigma^\mu \partial_\mu \psi_R &= 0 \end{aligned} \right\} \text{Weyl equations}$$

$$\gamma^5$$

If  $S[\Lambda]$  is not block diagonal, then is there an invariant way to define chiral spinors?

Introduce the "fifth" gamma-matrix  $\gamma^5 = i\gamma^0\gamma^1\gamma^2\gamma^3$

- NOTE:
- $\{\gamma^5, \gamma^\mu\} = 0$
  - $(\gamma^5)^2 = 1$
  - $[S_{\mu\nu}, \gamma^5] = 0 \rightarrow \gamma^5$  is a scalar under Lorentz transformation.

Define the Lorentz invariant projection operators:

$$P_{\pm} = \frac{1}{2}(1 \pm \gamma^5) \quad [ \quad P_+^2 = P_+, \quad P_-^2 = P_-, \\ P_+ P_- = 0 \quad ]$$

In the chiral representation:  $\gamma^5 = \begin{pmatrix} -1 & 0 \\ 0 & 1 \end{pmatrix} \rightarrow P_{\pm}$  project onto  $u_{\pm}$

For an arbitrary representation of the Clifford algebra, we use  $\gamma^5$  to define the chiral spinors.

$$\psi_{\pm} = P_{\pm} \psi \quad \left\{ \begin{array}{l} \psi_+ : \text{left-handed spinor} \\ \psi_- : \text{right-handed} \dots \end{array} \right.$$

## 5.2 Plane wave solutions [Schwartz 11.2]

Let's study the solutions to the Dirac equation

$$(i \gamma^\mu \partial_\mu - m) \psi(x) = 0.$$

Since  $\psi$  obeys the K-G equation, it has the form

$$\psi(x) = u(\vec{p}) e^{-i \vec{p} \cdot x} \quad [p^2 = m^2, \quad p^0 = \sqrt{m^2 + |\vec{p}|^2} > 0]$$

↑  
a four component spinor, only depends on  $\vec{p}$

Then the Dirac equation:

$$(i \gamma^\mu p_\mu - m) u(\vec{p}) = 0. \quad \Rightarrow \quad \begin{pmatrix} -m & \vec{p} \cdot \vec{\sigma} \\ \vec{p} \cdot \vec{\sigma} & -m \end{pmatrix} u(\vec{p}) = 0 \quad (*)$$

Then in the rest frame,  $p^\mu = (m, \vec{0}) \equiv p_0^\mu$

$$m \begin{pmatrix} -1 & 1 \\ 1 & -1 \end{pmatrix} u(p_0) = 0$$

$$\Rightarrow u(p_0) = \sqrt{m} \begin{pmatrix} \zeta \\ \zeta \end{pmatrix} \quad \zeta: \text{two-component spinor.}$$

e.g. two linearly independent solutions

$$\zeta_1 = \begin{pmatrix} 1 \\ 0 \end{pmatrix}, \quad \zeta_2 = \begin{pmatrix} 0 \\ 1 \end{pmatrix}$$

To derive a more general expression, if  $p^\mu = (E, 0, 0, p_z)$

$$\vec{p} \cdot \vec{\sigma} = \begin{pmatrix} E - p_z & 0 \\ 0 & E + p_z \end{pmatrix}; \quad \vec{p} \cdot \vec{\sigma} = \begin{pmatrix} E + p_z & 0 \\ 0 & E - p_z \end{pmatrix}$$

Let  $a = \sqrt{E - p_z}$ ,  $b = \sqrt{E + p_z}$ , then  $m^2 = a^2 b^2$ ,  $*$  becomes

$$\begin{pmatrix} -ab & 0 & a^2 & 0 \\ 0 & -ab & 0 & b^2 \\ b^2 & 0 & -ab & 0 \\ 0 & a^2 & 0 & -ab \end{pmatrix} u(p) = 0$$

The solutions are

$$u = \begin{pmatrix} \begin{pmatrix} a & 0 \\ 0 & b \end{pmatrix} \zeta \\ \begin{pmatrix} b & 0 \\ 0 & a \end{pmatrix} \zeta \end{pmatrix}$$

in the rest frame  $a^2 = b^2 = m^2$

in the  $p_z$  frame.

$$u_s(p) = \begin{pmatrix} \begin{pmatrix} \sqrt{E-p_z} & 0 \\ 0 & \sqrt{E+p_z} \end{pmatrix} \zeta_s \\ \begin{pmatrix} \sqrt{E+p_z} & 0 \\ 0 & \sqrt{E-p_z} \end{pmatrix} \zeta_s \end{pmatrix} \quad [s = 1, 2]$$

more generally,

$$u_s(p) = \begin{pmatrix} \sqrt{p \cdot \sigma} \zeta_s \\ \sqrt{p \cdot \bar{\sigma}} \zeta_s \end{pmatrix}$$

There are also solutions with  $p^0 < 0$ . [anti-particle]

$$\psi(x) = v(p) e^{i p \cdot x}$$

Similarly,

$$v_s(p) = \begin{pmatrix} \sqrt{p \cdot \sigma} \eta_s \\ -\sqrt{p \cdot \bar{\sigma}} \eta_s \end{pmatrix} \quad \eta_1 = \begin{pmatrix} 1 \\ 0 \end{pmatrix}, \quad \eta_2 = \begin{pmatrix} 0 \\ 1 \end{pmatrix}$$

Normalization and spin sums.

$$\begin{aligned} \bar{u}_s(p) u_{s'}(p) &= u_s^\dagger(p) \gamma^0 u_{s'}(p) = (\zeta_s^\dagger \sqrt{p \cdot \sigma}, \zeta_s^\dagger \sqrt{p \cdot \bar{\sigma}}) \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} \begin{pmatrix} \sqrt{p \cdot \sigma} \zeta_{s'} \\ \sqrt{p \cdot \bar{\sigma}} \zeta_{s'} \end{pmatrix} \\ &= (\zeta_s^\dagger, \zeta_s^\dagger) \begin{pmatrix} \sqrt{p \cdot \sigma} p \cdot \bar{\sigma} & 0 \\ 0 & \sqrt{p \cdot \bar{\sigma}} p \cdot \sigma \end{pmatrix} \begin{pmatrix} \zeta_{s'} \\ \zeta_{s'} \end{pmatrix} \\ &= 2m \delta_{ss'} \end{aligned}$$

$$\text{Similarly, } \bar{v}_s(p) v_{s'}(p) = -2m \delta_{ss'}$$

$$\bar{v}_s(p) u_{s'}(p) = \bar{u}_s(p) v_{s'}(p) = 0$$

The completeness relations.

$$\begin{aligned} \sum_{s=1,2} u_s(p) \bar{u}_s(p) &= \sum_s \begin{pmatrix} \sqrt{p \cdot \sigma} \zeta_s \\ \sqrt{p \cdot \bar{\sigma}} \zeta_s \end{pmatrix} (\zeta_s^\dagger \sqrt{p \cdot \sigma}, \zeta_s^\dagger \sqrt{p \cdot \bar{\sigma}}) \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} \\ &= \begin{pmatrix} m & p \cdot \sigma \\ p \cdot \bar{\sigma} & m \end{pmatrix} = \not{p} + m \\ &\quad \uparrow \\ &\quad \sum_{s=1,2} \zeta_s \zeta_s^\dagger = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} \end{aligned}$$

Similarly,

$$\sum_{s=1,2} v_s(p) \bar{v}_s(p) = \not{p} - m.$$

### 5.3 Quantization of the Dirac field. [ Peskin 3.5 ]

Impose the canonical commutation relations.

$$[\psi_a(t, \vec{x}), \psi_b^\dagger(t, \vec{y})] = \delta^{(3)}(\vec{x} - \vec{y}) \delta_{ab} \quad [i\psi^\dagger = \pi]$$

$a, b$  denote the spinor components.

$$[a_{\vec{p}}^r, a_{\vec{q}}^{s\dagger}] = [b_{\vec{p}}^r, b_{\vec{q}}^{s\dagger}] = (2\pi)^3 \delta^{(3)}(\vec{p} - \vec{q}) \delta^{rs}$$

We take

$$\psi(x) = \int \frac{d^3p}{(2\pi)^3} \frac{1}{\sqrt{2\omega_p}} \sum_{s=1}^2 [a_{\vec{p}}^s u^s(\vec{p}) e^{-i\vec{p}\cdot x} + b_{\vec{p}}^s v^s(\vec{p}) e^{i\vec{p}\cdot x}]$$

$u^s(\vec{p}) e^{-i\vec{p}\cdot \vec{x}}$  are eigenfunction of  $H$  with  $\omega_p$

$v^s(\vec{p}) e^{-i\vec{p}\cdot \vec{x}} \dots \dots \dots -\omega_p.$

$$\Rightarrow H = \int \frac{d^3p}{(2\pi)^3} \sum_s (\omega_p a_{\vec{p}}^{s\dagger} a_{\vec{p}}^s - \omega_p b_{\vec{p}}^{s\dagger} b_{\vec{p}}^s)$$

the negative-energy problem:  $\uparrow$  by creating more and more particles with  $b^\dagger$ , one can lower the energy indefinitely.

The general commutator:

$$[\psi_a(x), \bar{\psi}_b(y)] = (i\not{\partial}_x + m) [\phi(x), \phi(y)]$$

It vanishes as  $(x-y)^2 < 0$

Let  $|0\rangle$  be the state annihilated by  $a_{\vec{p}}^s$  and  $b_{\vec{p}}^s$ , i.e.  $a_{\vec{p}}^s |0\rangle = b_{\vec{p}}^s |0\rangle = 0$

$$\text{Then } [\psi_a(x), \bar{\psi}_b(y)] = \langle 0 | [\psi_a(x), \bar{\psi}_b(y)] | 0 \rangle$$

$$= \langle 0 | \psi_a(x) \bar{\psi}_b(y) | 0 \rangle - \langle 0 | \bar{\psi}_b(y) \psi_a(x) | 0 \rangle$$

||  
0



First consider  $\langle 0 | \psi(x) \bar{\psi}(y) | 0 \rangle$  : positive energy particle from  $y$  to  $x$

$\Rightarrow$  only  $a_p^{s\dagger}$  of  $\bar{\psi}(y)$  can contribute,  $b_p^{s\dagger}$  annihilate  $|0\rangle$

Similarly, in  $\langle 0 | \bar{\psi}(y) \psi(x) | 0 \rangle$ ,  $a_p^s$  still annihilates the vacuum, but  $b_p^s$  does not.

1°. Forget about the above commutation relations.

2°. Drop the definition of the vacuum  $|0\rangle$ .

To preserve causality, the spinor field anticommute. [P55-P56, Peskin]

$$\Rightarrow \langle 0 | \psi_a(x) \bar{\psi}_b(y) | 0 \rangle = - \langle 0 | \bar{\psi}_b(y) \psi_a(x) | 0 \rangle$$

$\Rightarrow$  Anticommutation relations for the Dirac field.

Equal-time commutation relations.

$$\{ \psi_a(t, \vec{x}), \psi_b^\dagger(t, \vec{y}) \} = \delta^{(3)}(\vec{x} - \vec{y}) \delta_{ab}$$

$$\{ \psi_a(t, \vec{x}), \psi_b(t, \vec{y}) \} = \{ \psi_a^\dagger(t, \vec{x}), \psi_b^\dagger(t, \vec{y}) \} = 0.$$

$$\Leftrightarrow \{ a_p^r, a_q^{s\dagger} \} = \{ b_p^r, b_q^{s\dagger} \} = (2\pi)^3 \delta^{(3)}(\vec{p} - \vec{q}) \delta^{rs}$$

The Hamiltonian,

$$H = \int \frac{d^3p}{(2\pi)^3} \frac{1}{2} ( \omega_p a_p^{s\dagger} a_p - \omega_p b_p^{s\dagger} b_p )$$

Simply redefine

$$\tilde{b}_p^s \equiv b_p^{s\dagger}, \quad \tilde{b}_p^{s\dagger} \equiv b_p^s$$

$$\Rightarrow N(H) = \int \frac{d^3p}{(2\pi)^3} \frac{1}{2} ( \omega_p a_p^{s\dagger} a_p + \omega_p \tilde{b}_p^{s\dagger} \tilde{b}_p^s )$$

If we choose  $|0\rangle$  to be the state annihilated by  $a_p^s$  and  $\tilde{b}_p^s$ , then all excitations of  $|0\rangle$  have positive energy.

Since  $(\tilde{b}^\dagger)^2 = 0$ , the state can not be filled twice.

$$a_p^\dagger a_q^\dagger |0\rangle = -a_q^\dagger a_p^\dagger |0\rangle$$

The ladder operators obey anticommutation relation  $\Rightarrow$  the particles obey Fermi-Dirac statistics.

Dirac introduced the concept of Dirac sea, which is that there is a full "sea" of occupied states of negative energy. Then  $b_p^\dagger$  destroys a particle of negative energy, which is equivalent to creating a "hole" in the Dirac sea, which acts as a state of positive energy.

Summary :

The field operators :

$$\psi(x) = \int \frac{d^3p}{(2\pi)^3} \frac{1}{\sqrt{2\omega_p}} \sum_s (a_p^s u^s(p) e^{-i\vec{p}\cdot\vec{x}} + b_p^{s\dagger} v^s(p) e^{i\vec{p}\cdot\vec{x}})$$

$$\bar{\psi}(x) = \int \frac{d^3p}{(2\pi)^3} \frac{1}{\sqrt{2\omega_p}} \sum_s (b_p^s \bar{v}^s(p) e^{-i\vec{p}\cdot\vec{x}} + a_p^{s\dagger} \bar{u}^s(p) e^{i\vec{p}\cdot\vec{x}})$$

The creation and annihilation operators.

$$\{a_p^r, a_q^{s\dagger}\} = \{b_p^r, b_q^{s\dagger}\} = (2\pi)^3 \delta^{(3)}(\vec{p}-\vec{q}) \delta^{rs} \quad ; \quad \text{others} = 0$$

The equal-time anticommutation

$$\{\psi_a(t, \vec{x}), \psi_b^\dagger(t, \vec{y})\} = \delta_{ab} \delta^{(3)}(\vec{x}-\vec{y})$$

$$\text{others} = 0$$

The vacuum  $|0\rangle$  :  $a_p^s |0\rangle = b_p^s |0\rangle = 0$ .

The Hamiltonian  $N(H) = \int \frac{d^3p}{(2\pi)^3} \sum_s (\omega_p a_p^{s\dagger} a_p + \omega_p b_p^{s\dagger} b_p)$

The momentum  $N(\vec{P}) = \int \frac{d^3p}{(2\pi)^3} \sum_s (\vec{p} a_p^{s\dagger} a_p + \vec{p} b_p^{s\dagger} b_p)$

Both  $a_p^{s\dagger}$  and  $b_p^{s\dagger}$  create particles with  $\omega_p$ ,  $\vec{p}$

$a_p^{s\dagger}$  : fermion

$b_p^{s\dagger}$  : anti-fermion.

The one particle states  $|\vec{p}, s\rangle = \sqrt{2\omega_p} a_p^{s\dagger} |0\rangle$

Feynman propagator.

$$S_F(x-y) \equiv \langle 0 | T \psi(x) \bar{\psi}(y) | 0 \rangle$$

$$= \theta(x^0 - y^0) \langle 0 | \psi(x) \bar{\psi}(y) | 0 \rangle - \theta(y^0 - x^0) \langle 0 | \bar{\psi}(y) \psi(x) | 0 \rangle$$

where

$$\langle 0 | \psi_a(x) \bar{\psi}_b(y) | 0 \rangle = \int \frac{d^3 p}{(2\pi)^3} \frac{1}{2\omega_p} \sum_s u_a^s(p) \bar{u}_b^s(p) e^{-i p \cdot (x-y)}$$

$$= (i \not{\partial}_x + m)_{ab} D(x-y)$$

$$\langle 0 | \bar{\psi}_b(y) \psi_a(x) | 0 \rangle = -(i \not{\partial}_x + m)_{ab} D(x-y)$$

Then

$$S_F(x-y) = (i \not{\partial}_x + m) D_F(x-y)$$

$$= \int \frac{d^4 p}{(2\pi)^4} \frac{i(\not{p} + m)}{p^2 - m^2 + i\epsilon} e^{-i p \cdot (x-y)}$$

#### 5.4 Feynman rules for fermions. [Peskin 4.7]

The time-ordered product picks up  $(-1)$  for each interchange of operators.

$$T(\psi_1 \psi_2 \psi_3 \psi_4) = (-1)^3 \psi_3 \psi_1 \psi_4 \psi_2 \quad \text{for } x_3^0 > x_1^0 > x_4^0 > x_2^0$$

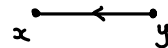
Similarly, the normal order product.

$$N(a_p a_q a_r^\dagger) = (-1)^3 a_r^\dagger a_p a_q = (-1)^3 a_r^\dagger a_q a_p.$$

The Wick's theorem for two operators:

$$T[\psi(x) \bar{\psi}(y)] = N[\psi(x) \bar{\psi}(y)] + \overbrace{\psi(x) \bar{\psi}(y)}$$

$$\text{with } \overbrace{\psi(x) \bar{\psi}(y)} = S_F(x-y)$$



$$\overbrace{\psi(x) \psi(y)} = \overbrace{\bar{\psi}(x) \bar{\psi}(y)} = 0.$$

Define contractions under the normal order as

$$N(\overbrace{\psi_1 \psi_2 \bar{\psi}_3 \bar{\psi}_4}) = -\overbrace{\psi_1 \bar{\psi}_3} N(\psi_2 \bar{\psi}_4) = -S_F(x_1 - x_3) N(\psi_2 \bar{\psi}_4)$$

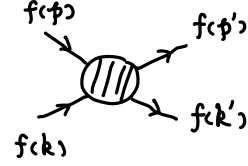
The Wick's theorem.

$$T[\psi_1 \bar{\psi}_2 \psi_3 \dots] = N[\psi_1 \bar{\psi}_2 \psi_3 \dots + \text{all possible contractions}]$$

## 5.5 Yukawa theory. [ Peskin 4.7 ]

$$\mathcal{L}_{\text{Yukawa}} = \mathcal{L}_{\text{Dirac}} + \mathcal{L}_{\text{Klein-Gordon}} - g\bar{\psi}\psi\phi.$$

$$H = H_{\text{Dirac}} + H_{\text{Klein-Gordon}} + \int d^3x g\bar{\psi}\psi\phi.$$



To be even more specific, consider the two-particle scattering reaction

$$\text{fermion}(p) + \text{fermion}(k) \longrightarrow \text{fermion}(p') + \text{fermion}(k').$$

The leading contribution comes from the  $H_I^2$  term of the  $S$ -matrix:

$$\langle \mathbf{p}', \mathbf{k}' | T \left( \frac{1}{2!} (-ig) \int d^4x \bar{\psi}_I \psi_I \phi_I \quad (-ig) \int d^4y \bar{\psi}_I \psi_I \phi_I \right) | \mathbf{p}, \mathbf{k} \rangle_0. \quad (4.113)$$

To evaluate this expression, use Wick's theorem to reduce the  $T$ -product to an  $N$ -product of contractions, then act the uncontracted fields on the initial- and final-state particles. Represent this latter process as the contraction

$$\begin{aligned} \overline{\psi_I(x) | \mathbf{p}, s} &= \int \frac{d^3p'}{(2\pi)^3} \frac{1}{\sqrt{2E_{\mathbf{p}'}}} \sum_{s'} a_{\mathbf{p}'}^{s'} u^{s'}(p') e^{-ip' \cdot x} \sqrt{2E_{\mathbf{p}}} a_{\mathbf{p}}^{s\dagger} | 0 \rangle \\ &= e^{-ip \cdot x} u^s(p) | 0 \rangle. \end{aligned} \quad (4.114)$$

Similar expressions hold for the contraction of  $\bar{\psi}_I$  with a final-state fermion, and for contractions of  $\psi_I$  and  $\bar{\psi}_I$  with antifermion states. Note that  $\psi_I$  can be contracted with a fermion on the right or an antifermion on the left; the opposite is true for  $\bar{\psi}_I$ .

We can write a typical contribution to the matrix element (4.113) as the contraction

$$\overline{\langle \mathbf{p}', \mathbf{k}' | \frac{1}{2!} (-ig) \int d^4x \bar{\psi}\psi\phi \quad (-ig) \int d^4y \bar{\psi}\psi\phi | \mathbf{p}, \mathbf{k} \rangle}. \quad (4.115)$$

Up to a possible minus sign, the value of this quantity is

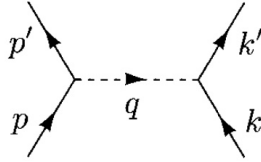
$$\begin{aligned} &(-ig)^2 \int \frac{d^4q}{(2\pi)^4} \frac{i}{q^2 - m_\phi^2} (2\pi)^4 \delta(p' - p - q) \\ &\quad \times (2\pi)^4 \delta(k' - k + q) \bar{u}(p') u(p) \bar{u}(k') u(k). \end{aligned}$$

(We have dropped the factor  $1/2!$  because there is a second, identical term that comes from interchanging  $x$  and  $y$  in (4.115).) Using either delta function to perform the integral, we find that this expression takes the form  $i\mathcal{M}(2\pi)^4 \delta(\Sigma p)$ , with

$$i\mathcal{M} = \frac{-ig^2}{q^2 - m_\phi^2} \bar{u}(p') u(p) \bar{u}(k') u(k). \quad (4.116)$$

When writing it in this way, we must remember to impose the constraints  $p - p' = q = k' - k$ .

Instead of working from (4.115), we could draw a Feynman diagram:



We denote scalar particles by dashed lines, and fermions by solid lines. The  $S$ -matrix element could then be obtained directly from the following momentum-space Feynman rules.

1. Propagators:

$$\overline{\phi(x)\phi(y)} = \text{---} \xrightarrow{q} \text{---} = \frac{i}{q^2 - m_\phi^2 + i\epsilon}$$

$$\overline{\psi(x)\psi(y)} = \text{---} \xrightarrow{p} \text{---} = \frac{i(\not{p} + m)}{p^2 - m^2 + i\epsilon}$$

2. Vertices:

$$\text{---} \text{---} \text{---} = -ig$$

3. External leg contractions:

$$\overline{\phi|\mathbf{q}} = \text{---} \text{---} \text{---} = 1 \quad \langle \mathbf{q}|\phi = \text{---} \text{---} \text{---} = 1$$

$$\underbrace{\overline{\psi|\mathbf{p}, s}}_{\text{fermion}} = \text{---} \text{---} \text{---} = u^s(p) \quad \underbrace{\langle \mathbf{p}, s|\bar{\psi}}_{\text{fermion}} = \text{---} \text{---} \text{---} = \bar{u}^s(p)$$

$$\underbrace{\overline{\psi|\mathbf{k}, s}}_{\text{antifermion}} = \text{---} \text{---} \text{---} = \bar{v}^s(k) \quad \underbrace{\langle \mathbf{k}, s|\psi}_{\text{antifermion}} = \text{---} \text{---} \text{---} = v^s(k)$$

4. Impose momentum conservation at each vertex.
5. Integrate over each undetermined loop momentum.
6. Figure out the overall sign of the diagram.

Finally, let's take a moment to worry about fermion minus signs. Return to the example of the fermion-fermion scattering process. We adopt a sign convention for the initial and final states:

$$|\mathbf{p}, \mathbf{k}\rangle \sim a_{\mathbf{p}}^{\dagger} a_{\mathbf{k}}^{\dagger} |0\rangle, \quad \langle \mathbf{p}', \mathbf{k}'| \sim \langle 0| a_{\mathbf{k}'} a_{\mathbf{p}'}, \quad (4.118)$$

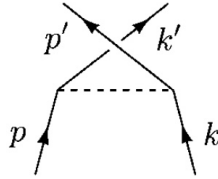
so that  $(|p, k\rangle)^{\dagger} = \langle p, k|$ . Then the contraction

$$\langle \mathbf{p}', \mathbf{k}' | (\bar{\psi}\psi)_x (\bar{\psi}\psi)_y | \mathbf{p}, \mathbf{k} \rangle \sim \langle 0 | a_{\mathbf{k}'} a_{\mathbf{p}'} \bar{\psi}_x \psi_x \bar{\psi}_y \psi_y a_{\mathbf{p}}^{\dagger} a_{\mathbf{k}}^{\dagger} | 0 \rangle$$

can be untangled by moving  $\bar{\psi}_y$  two spaces to the left, and so picks up a factor of  $(-1)^2 = +1$ . But note that in the contraction

$$\langle \mathbf{p}', \mathbf{k}' | (\bar{\psi}\psi)_x (\bar{\psi}\psi)_y | \mathbf{p}, \mathbf{k} \rangle \sim \langle 0 | a_{\mathbf{k}'} a_{\mathbf{p}'} \bar{\psi}_x \psi_x \bar{\psi}_y \psi_y a_{\mathbf{p}}^{\dagger} a_{\mathbf{k}}^{\dagger} | 0 \rangle,$$

it is sufficient to move the  $\bar{\psi}_y$  one space to the left, giving a factor of  $-1$ . This contraction corresponds to the diagram



The full result, to lowest order, for the  $S$ -matrix element for this process is therefore

$$\begin{aligned} i\mathcal{M} &= \text{Diagram 1} + \text{Diagram 2} \\ &= (-ig^2) \left( \bar{u}(p') u(p) \frac{1}{(p' - p)^2 - m_{\phi}^2} \bar{u}(k') u(k) \right. \\ &\quad \left. - \bar{u}(p') u(k) \frac{1}{(p' - k)^2 - m_{\phi}^2} \bar{u}(k') u(p) \right). \end{aligned} \quad (4.119)$$

The minus sign difference between these diagrams is a reflection of Fermi statistics. Turning this expression into an explicit cross section would require